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FREQUENCY SELECTIVE STRUCTURE WITH ARTIFICIAL DIELECTRIC MATCHING ELEMENTS

Abstract

A frequency selective structure featuring one or more layers of artificial dielectric matching elements. The artificial dielectric matching elements have an open distal end and a closed base end. The closed base end of each artificial dielectric matching element is disposed on or adjacent to a first surface of a substrate. Artificial dielectric matching elements can also be disposed on or adjacent a second surface of the substrate. The substrate can be configured as a frequency selective surface. The artificial dielectric matching elements can have interior volumes that are wider adjacent to the open distal end, and narrower adjacent to the closed base end. Any non-conductive material can be used to form the artificial dielectric matching elements. The artificial dielectric matching elements can be placed using automated pick and place operations.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/563,501, filed Mar. 11, 2024, the entire disclosure of which is hereby incorporated herein by reference.

FIELD

[0002] Systems and methods for providing a frequency selective surface or structure incorporating an artificial dielectric that enables the transmission of electromagnetic waves within a selected range of frequencies through a structure at a wide range of relative angles are provided.

BACKGROUND

[0003] Antennas are often used in environments in which it is desirable to provide a cover or radome to protect the antenna and to provide a surface that conforms to the vehicle or other structure to which the antenna is interconnected. In addition to providing adequate structural integrity and durability for a given set of operating conditions, a radome must be transparent to electromagnetic energy within the intended operating wavelengths of the associated antenna. In addition, it can be desirable to provide a radome that rejects wavelengths outside of the intended operating range. Moreover, it can be desirable to provide a radome that provides selected wavelength transparency to signals having multiple polarizations. In order to provide such characteristics, a radome often incorporates a frequency selective surface.

[0004] A frequency selective surface is a type of structure that passes electromagnetic energy within a selected wavelength range and that stops or blocks electromagnetic energy outside of that selected wavelength range. Although such structures can be designed with great precision with respect to signals that are normal (or at some other predetermined angle) to the surface of the structure, the effective passband presented to the electromagnetic energy changes with the angle of incidence. Accordingly, the performance of an antenna can be compromised along certain relative lines of sight by a radome that features curves or where the antenna covered by the radome is mechanically or electronically scanned. The adverse effects of a radome on antenna performance can in some cases be reduced by providing additional layers of material adjacent to the frequency selective surface. In particular, the inclusion of layers of relatively low dielectric constant material can extend the range of angles over which a desired wavelength response is obtained. However, it can be difficult or impossible to obtain a material having the required dielectric constant. In addition, materials having low dielectric constants are often foams or other porous, mechanically weak materials. Such materials can also be prone to absorbing water and other atmospheric constituents, and can have coefficients of thermal expansion that are not well matched to that of other materials in the structure.

[0005] Accordingly, it would be desirable to provide a frequency selective structure that enabled selected transmission characteristics to electromagnetic energy of any polarization over a wide range of angles, and that could be constructed without requiring materials having relatively low dielectric constants.

SUMMARY

[0006] Embodiments of the present disclosure are directed to providing frequency selective structures with favorable transmission characteristics with respect to radio frequency energy over a range of incident angles. A frequency selective structure as disclosed herein generally includes a substrate and one or more artificial dielectric matching layers that each include an array of artificial

dielectric matching elements. The artificial dielectric matching elements can be formed from a dielectric material and can be configured as cylinders or faceted elements with a closed base end adjacent the substrate and an open distal end opposite the base end, where adjacent is close to or on. An interior volume of each artificial dielectric matching element can include more than one interior volume portion. For example, a first interior volume portion having a first diameter can be disposed adjacent the base, and a second interior volume portion having a second diameter, where the second diameter is larger than the first diameter, can be disposed adjacent the open distal end. A coating can be disposed over the artificial dielectric matching elements, filling the interior volumes and the spaces between adjacent elements, to form a smooth surface.

[0007] In accordance with at least some embodiments of the present disclosure, a frequency selective structure can include a frequency selective surface. For instance, a frequency selective surface can be disposed between the substrate and an artificial dielectric matching layer having an array of artificial dielectric matching elements. As another example, the substrate can be configured as a thick frequency selective surface, with through holes in which paired artificial dielectric matching elements are received.

[0008] Frequency selective structures in accordance with embodiments of the present disclosure provide selected wavelength transparency to electromagnetic energy over a wide range of angles. In addition, the frequency selective structures can provide selected wavelength transparency to electromagnetic energy having any of multiple different polarizations.

[0009] Additional features and advantages of embodiments of the present disclosure will become more readily apparent from the following description, particularly when taken together with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIGS. 1A and 1B illustrate example antennas and associated radomes that incorporate a frequency selective structure in accordance with embodiments of the present disclosure;

[0011] FIG. 2 is a cross section in elevation of a frequency selective structure incorporating a frequency selective surface in accordance with the prior art;

[0012] FIG. 3 is a plan view of a frequency selective structure incorporating a frequency selective surface in accordance with the prior art;

[0013] FIG. 4 is a graph depicting the transmission performance of an example frequency selective structure in accordance with the prior art;

[0014] FIG. 5 is a cross section in elevation of a frequency selective structure incorporating a frequency selective surface and dielectric matching layers in accordance with the prior art;

[0015] FIG. 6A is a cross section in elevation of a frequency selective structure incorporating a frequency selective surface and artificial dielectric matching elements in accordance with embodiments of the present disclosure;

[0016] FIG. 6B is a cross section in elevation of a frequency selective structure incorporating a frequency selective surface and artificial dielectric matching elements in accordance with other embodiments of the present disclosure;

[0017] FIG. 7 is a plan view of a frequency selective structure incorporating artificial dielectric matching elements in accordance with embodiments of the present disclosure;

[0018] FIG. 8 is a plan view of a frequency selective surface of a frequency selective structure in accordance with embodiments of the present disclosure;

[0019] FIG. 9 is a cross-section in elevation of an artificial dielectric matching element in accordance with embodiments of the present disclosure;

[0020] FIG. 10 is a top plan view of an artificial dielectric matching element in accordance with

embodiments of the present disclosure;

[0021] FIG. **11A** is a cross section in elevation of a unitary frequency selective structure incorporating artificial dielectric matching elements in accordance with embodiments of the present disclosure;

[0022] FIG. **11B** is a cross section in elevation of a unitary frequency selective structure incorporating artificial dielectric matching elements in accordance with other embodiments of the present disclosure;

[0023] FIG. **12** is a plan view of a unitary frequency selective structure incorporating artificial dielectric matching elements in accordance with embodiments of the present disclosure;

[0024] FIG. **13A** is a cross section in elevation of a frequency selective structure incorporating a thick frequency selective surface and artificial dielectric matching elements in accordance with embodiments of the present disclosure;

[0025] FIG. **13B** is a cross section in elevation of a frequency selective structure incorporating a thick frequency selective surface and artificial dielectric matching elements in accordance with other embodiments of the present disclosure;

[0026] FIG. **14** is a plan view of a frequency selective structure incorporating a thick frequency selective surface and artificial dielectric matching elements in accordance with embodiments of the present disclosure;

[0027] FIG. **15** is a graph depicting the transmission performance of a frequency selective structure incorporating artificial dielectric matching elements in accordance with embodiments of the present disclosure; and

[0028] FIG. **16** is a flow chart illustrating aspects of a method for forming a frequency selective structure incorporating an artificial dielectric matching layer having a plurality of artificial dielectric matching elements in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0029] Embodiments of the present disclosure are generally directed to providing frequency selective structures that provide selected transmission characteristics for electromagnetic energy or waves over a wide range of angles. More particularly, embodiments of the present disclosure are directed to frequency selective structures that incorporate an artificial dielectric matching layer having an array of artificial dielectric matching structures or elements to provide desired transmission characteristics over a wide range of angles. In accordance with further embodiments of the present disclosure, frequency selective structures are provided that attenuate or block electromagnetic energy within a selected stop band over a wide range of angles. In accordance with still further embodiments, a frequency selective structure incorporating an artificial dielectric matching layer comprising a plurality of artificial dielectric matching elements as disclosed herein provides desired transmission characteristics to electromagnetic energy having any of multiple different polarization states. Frequency selective structures as disclosed herein are particularly well suited for use in radomes or antenna covers, or on any other structure disposed directly adjacent or near a transmitting or receiving antenna element for which specified electromagnetic energy transmission characteristics are desired or required.

[0030] FIGS. **1A** and **1B** illustrate example antennas and associated radomes that incorporate a frequency selective structure in accordance with embodiments of the present disclosure. In FIG. **1A**, an antenna **104**, such as a weather radar, target acquisition radar, or radio communication antenna disposed on a structure **108**, in this example an aircraft, is depicted. The antenna **104** is disposed within an interior volume of an antenna cover or radome **112**, shown in cross-section in the figure. The radome **112** can form part of the aerodynamic structure of the aircraft, and serves to protect the antenna **104** from the surrounding environment. In FIG. **1B**, a plurality of antennas **104**, such as a set of cellular telephony antennas interconnected to a structure **108**, in this example a cellular telephone tower, are depicted. The antennas **104** are disposed within an interior volume of an antenna cover or radome **112**, shown in cross-section in the figure, which provides protection for

the antennas **104** from the surrounding environment. As can be appreciated by one of skill in the art, it is generally desirable to provide a radome **112** that allows the operating frequencies of an associated antenna **104** to pass through the radome **112** without attenuation or other modification of a transmitted or received signal. In addition, it can be desirable to provide a radome **112** that blocks radio frequency energy within selected other frequency ranges. Accordingly, an antenna cover or radome **112** is typically configured as a frequency selective structure. However, providing a radome **112** that provides desired transmission characteristics to radio frequency energy that is incident on a surface of the radome **112** at different angles, for example because the radome **112** is curved or because a beam is mechanically or electronically scanned, has been difficult.

[0031] FIG. 2 is a cross section in elevation and FIG. 3 is a plan view of a basic frequency selective structure **204** in accordance with the prior art. The frequency selective structure **204** can include first **208a** and second **208b** frequency selective surfaces or layers spaced apart from one another by a dielectric layer **216**. The frequency selective surfaces **208** are resonant repeating structures that are disposed directly on opposite surfaces of the dielectric layer **216**. The dielectric layer **216** can be formed from a material having a relatively low dielectric constant. The thickness of the dielectric layer **216**, and thus the spacing between the two frequency selective surfaces **208a** and **208b**, can be some fraction of a wavelength of a signal to be passed by the frequency selective structure **204**. The frequency selective surfaces **208a** and **208b** can be configured identically, and can each include an electrically conductive material having a plurality of unit cells **220** disposed in an array, with each unit cell **220** containing a slot element **224** formed as an open or nonconductive area (see FIG. 3). The size and configuration of the slot elements **224** of the unit cells **220** is dependent on the desired transmission characteristics of the frequency selective structure **204**.

[0032] As illustrated in FIG. 4, the transmission performance of a frequency selective structure **204** with respect to electromagnetic energy at a normal angle to the frequency selective structure **204** is not well aligned with the transmission performance of the frequency selective structure **204** with respect to electromagnetic energy at non-normal angles. Accordingly, it is difficult to provide a frequency selective structure **204** that provides transparency to electromagnetic energy having a wavelength within a selected band while rejecting electromagnetic energy at wavelengths outside of that selected band over a wide range of angles.

[0033] In order to improve the off normal performance of a basic prior art frequency selective structure **204**, and as illustrated in FIG. 5, a modified frequency selective structure **504** can include additional layers of dielectric material layers **508a** and **508b** disposed on either side of the basic frequency selective structure **204**. For instance, the dielectric material layers **508** can be disposed directly on surfaces of the frequency selective surfaces **208** of the basic frequency selective structure **204**. Provided each of the additional layers of dielectric material **508** feature a low dielectric constant (e.g. about 1.2), the modified frequency selective structure **504** can exhibit transmission performance that is relatively consistent over a range of angles. Accordingly, the normal angle and off-normal angle performance of the modified frequency selective structure **504** can be improved as compared to that of the basic frequency selective structure **204**. However, it can be difficult to identify a material with the required dielectric constant. Moreover, materials exhibiting a sufficiently low dielectric constant are often foams or other porous materials that can have poor structural characteristics and that can absorb moisture or other atmospheric constituents.

[0034] A cross-section in elevation of a portion of a frequency selective structure **604** in accordance with embodiments of the present disclosure is depicted in FIG. 6A. The frequency selective structure **604** features one or more artificial dielectric matching layers **608** as disclosed herein. In accordance with at least some embodiments, the frequency selective structure **604** includes first **612a** and second **612b** frequency selective surface layers spaced apart from one another by a substrate **616**. In this example, the substrate **616** is a dielectric layer. The thickness of the substrate **616**, and thus the distance between the first **612a** and second **612b** frequency selective surface layers, is a fraction of a wavelength to which the frequency selective structure **604** is transparent. In

addition, the size and spacing of the unit cells **808** (see FIGS. **6A** and **8**) of the frequency selective surface layers **612** can be based on the desired transmission performance of the frequency selective structure **604**, and can be determined using the same or similar techniques as would be used to design a basic **204** or modified **504** frequency selective structure of the prior art. Accordingly, certain elements of a frequency selective structure **604** can have a configuration that is the same as or similar to the basic prior art frequency selective structure **204** illustrated in FIGS. **2** and **3** and the modified frequency selective structure **504** illustrated in FIG. **5**. However, rather than including the layers **508** of low dielectric material as in the modified frequency selective structure **504**, a frequency selective structure **604** in accordance with embodiments of the present disclosure includes one or more of artificial dielectric matching layers **608**.

[0035] As shown in FIG. **6A**, a first artificial dielectric matching layer **608a** that includes a first set of artificial dielectric matching elements **620** is disposed on or adjacent a surface of the first frequency selective surface layer **612a**, and a second artificial dielectric matching layer **608b** that includes a second set of artificial dielectric matching elements **620** is disposed on or adjacent a surface of the second frequency selective surface layer **612b**. The artificial dielectric matching elements **620** included in the artificial dielectric matching layers **608** can be disposed directly on or adjacent to a corresponding one of the frequency selective surface layers **612**. Accordingly, the artificial dielectric matching layers **608** can form the outer layers of the frequency selective structure **604**. In accordance with further embodiments of the present disclosure, and as illustrated in FIG. **6B**, which is otherwise the same as the embodiment illustrated in FIG. **6A**, the frequency selective structure **604** can also include a coating **624** that covers the artificial dielectric matching elements **620** of one or both of the artificial dielectric matching layers **608**, filling the spaces between the artificial dielectric matching elements **620** and interior volumes **913** (see FIG. **9**) of the artificial dielectric matching elements **620**, thereby creating a smooth surface. The coating **624** can include a dielectric material. As an example, but without limitation, the coating **624** can include a low density ablator or insulator in the form of a filled elastomeric silicone commonly used for thermal protection on aircraft. The coating **624** can also be in the form of a white silicon based paint or thick film coating or an epoxy coating.

[0036] As previously noted, each artificial dielectric matching layer **608** includes a plurality of artificial dielectric matching elements **620**. As illustrated in FIG. **7**, which is a plan view of a frequency selective structure **604** in accordance with embodiments of the present disclosure, the artificial dielectric matching elements **620** in an artificial dielectric matching layer **608a** or **608b** (labeled simply as **608** in FIG. **7**) can be disposed in an array **704**. The array can be a two dimensional array or lattice. For instance, as depicted in FIG. **7**, the lattice can be a rectangular lattice. However, other patterns or dispositions are possible. For instance, the lattice can be a hexagonal lattice. In addition, the artificial dielectric matching elements **620** within an artificial dielectric matching layer **608** can all have the same composition and size, and can be equally spaced from one another. The spacing and configuration of the artificial dielectric matching elements **620** in an artificial dielectric matching layer **608** can correspond to the spacing and configuration of the unit cells **808** of the immediately adjacent frequency selective surface layer **612** (see FIGS. **6A**, **6B**, and **8**). For instance, the artificial dielectric matching elements **620** and the unit cells **808** can be disposed in matching two-dimensional arrays, such that each artificial dielectric matching element **620** is within an area of a corresponding unit cell **808**. In accordance with embodiments of the present disclosure, the spacing and configuration of the arrays of artificial dielectric matching elements **620** of the artificial dielectric matching layers **608** within a frequency selective structure **604** can be the same or different from one another.

[0037] FIG. **8** is a plan view of a frequency selective surface layer **612** of a frequency selective structure **604** in accordance with embodiments of the present disclosure. The frequency selective surface layer **612** can be configured as an electrically conductive material **804** having a plurality of unit cells **808** disposed in an array **812**. Each of the unit cells **808** includes a slot element **816**,

formed as an open or otherwise nonconductive area in the conductive material **804**. The size and shape of the slot elements **816** in the individual unit cells **808**, and the spacing between the unit cells **808** of the array **812**, can be determined based on the desired passband and stop band characteristics of the frequency selective structure **604**.

[0038] In accordance with embodiments of the present disclosure, each artificial dielectric matching element **620** of an artificial dielectric matching layer **608** is associated with a corresponding unit cell **808** of an immediately adjacent frequency selective surface layer **612**. Moreover, each artificial dielectric matching element **620** is disposed on or adjacent a side of a frequency selective surface layer **612** opposite the side of that frequency selective surface layer **612** that is on or facing the substrate **616**. For example, each artificial dielectric matching element **620** can be disposed in an area corresponding to an area of a unit cell **808** of the frequency selective surface layer **612**. An exterior diameter of the dielectric matching elements **620** can be less than a width or length of the unit cells **808**. In accordance with at least some embodiments of the present disclosure, each artificial dielectric matching element **620** is centered within an area of a corresponding unit cell **808**. Therefore, comparing the array **704** of artificial dielectric matching elements **620** illustrated in FIG. 7 to the array **812** of unit cells **808** illustrated in FIG. 8, it can be seen that the configuration and spacing of the artificial dielectric matching elements **620** matches that of the unit cells **808**. As an example, both the artificial dielectric matching elements **620** and the unit cells **808** can be disposed in an array in the form of a regular lattice.

[0039] FIG. 9 is a cross-section in elevation of an artificial dielectric matching element **620** and FIG. 10 is a top plan view of an artificial dielectric matching element **620** in accordance with embodiments of the present disclosure. In FIG. 9, only surfaces along the section are shown for clarity. As illustrated, each artificial dielectric matching element **620** can be configured as a cylindrical element with a closed base portion **905** that is adjacent or directly on a corresponding unit cell **808**, and an open end or distal portion **909** opposite the base portion **905**. Accordingly, in a frequency selective structure **604** as described herein, an artificial dielectric matching element **620** has an interior void or volume **913** that extends between an open end **909**, which faces away from an adjacent frequency selective surface layer **612**, and a closed end **921** at or coincident with the base portion **905**. In accordance with at least some embodiments of the present disclosure, an exterior diameter **923** of the artificial dielectric matching element **620** is constant along a length of the element **620**, from the base portion **905** to the end **909** opposite to the base portion. In addition, a diameter of the interior volume **913** can be different at different distances from the closed base portion **905**. Accordingly, an artificial dielectric matching element **620** can include multiple interior volume portions corresponding to areas within the interior volume **913** having different interior diameters.

[0040] For instance, and as illustrated in FIGS. 9 and 10, a first interior volume portion **917** of the interior volume **913** having a first diameter **915** is disposed adjacent the closed end **921** of the interior volume **913**, and a second interior volume portion **925** having a second diameter **927**, where the second diameter is larger than the first diameter, is disposed between a shoulder surface **929**, formed at the interface of the first diameter **915** and second diameter **927** portions, and the open end **909**. Accordingly, the first interior volume portion **917** and the second interior volume portion **925** can be in the form of cylindrical volumes that are concentric with one another and with the exterior diameter **923** of the artificial dielectric matching element **620**. Although an example with two interior volume portions **917** and **925** has been illustrated, it should be appreciated that an artificial dielectric matching element **620** is not limited to any particular number of interior volume portions. In accordance with other embodiments of the present disclosure, an artificial dielectric matching element **620** can include an interior volume or void **913** with a single, constant diameter interior volume portion. In accordance with still other embodiments of the present disclosure, the interior volume **913** of an artificial dielectric matching element **620** can have a diameter that increases with distance from the base portion **905**. Where the diameter of the interior volume **913**

increases with distance from the base portion **905**, the increase can be linear, exponential, stepped, or according to some other function. Artificial dielectric matching elements **620** in accordance with still further embodiments of the present disclosure can also include faceted outer surfaces and/or a faceted interior volumes **913**. As can be appreciated by one of skill in the art after consideration of the present disclosure, an artificial dielectric matching element **620** can therefore include more dielectric material at or near the base portion **905** than at the open distal portion or end **909**.

Moreover, the exterior diameter **923** can remain constant, with the effect that the walls of the artificial dielectric matching elements are thinner at the open end **909** than adjacent the base **905**. [0041] In accordance with embodiments of the present disclosure, the artificial dielectric matching elements **620** can be formed from any insulator. Examples of suitable materials include standard circuit board materials, glass, fused silica, and ceramics. Moreover, the material from which the dielectric matching elements **620** are formed need not have a particularly low dielectric constant. As such, the material of the artificial dielectric matching elements **620** can have a dielectric constant of anywhere from 1.3 to 9.0. As a further example, the material of the artificial dielectric matching elements **620** can have a dielectric constant of from 1.4 to 4.0.

[0042] The size of the artificial dielectric matching elements **620**, the size of the interior volume **913** and the sizes of any included interior volume portions **917** or **925** can be tuned to match the low angle response of the resulting frequency selective structure **604** to the normal angle response of that structure **604**. As examples, but without limitation, the diameter **923** of the exterior cylindrical surface of each artificial dielectric matching element **620** can be about one-half a wavelength of a selected signal (and the size of each slot element **816** of the frequency selective surface layer **612** can likewise be about one-half a wavelength of the selected signal). The height **931** of the artificial dielectric matching element **620**, from the base **905** to a top or distal edge **911**, can be between one-quarter and one-half a wavelength of the selected signal. The first interior volume portion **917** can have a diameter that is between 50% and 70% of the outer diameter **923** of the artificial dielectric matching element **620** and can extend to a point that is about 90% of a distance from the open distal end **909** to an outer surface of the base portion **905**. The second interior volume portion **925** can have a diameter that is between 75% and 95% of the outer diameter **923** of the artificial dielectric matching element **620** and can extend from the open distal end **909** to about 60% of the way to the base portion **905**.

[0043] FIG. **11A** is a cross section in elevation and FIG. **12** is a plan view of a one-piece or unitary frequency selective structure **1104** incorporating artificial dielectric matching layers **1108** that each include an array of artificial dielectric matching elements **1120** that are integral to a substrate **1116** in accordance with embodiments of the present disclosure. More particularly, the artificial dielectric matching elements **1120** and the substrate **1116** are formed from the same piece of material. The material may have a dielectric constant ranging from 1.3 to 9.0. As an example, the unitary frequency selective structure **1104** can be molded from a plastic polycarbonate. As another example, the unitary frequency selective structure **1104** can be three-dimensional printed using polylactic acid (PLA) or acrylonitrile butadiene styrene (ABS). As illustrated in FIG. **11B**, which is otherwise the same as FIG. **11A**, a coating **1124** can be applied over the artificial dielectric matching elements **1120** of one or both of the artificial dielectric matching layers **1108a** and **1108b**, filling the interior volumes **1113** of the artificial dielectric matching elements **1120** and the spaces between neighboring artificial dielectric matching elements **1120**, creating a smooth surface. As in other embodiments, examples of the coating **1124** can include a dielectric material such as a low density ablator or insulator in the form of a filled elastomeric silicone, a white silicon based paint, a thick film coating, or an epoxy coating.

[0044] In the embodiments of FIGS. **11A**, **11B**, and **12**, a metallized frequency selective surface is not included. Accordingly, in a unitary frequency selective structure **1104** as disclosed herein, the size, configuration and spacing of the artificial dielectric matching elements **1120**, together with the material selected to form the structure **1104**, enables selected performance characteristics to be

obtained. The artificial dielectric matching elements **1120** included in a unitary frequency selective structure **1104** as disclosed herein can include a base portion **1105** defining a closed end **1121** of an interior volume **1113**, a first interior volume portion **1117** having a first diameter **1115**, and a second interior volume portion **1125** having a second diameter **1127** that is larger than the first diameter **1115**. The material selected for forming the unitary frequency selective structure **1104**, the dimensions of the artificial dielectric matching elements **1120**, the spacing of the artificial dielectric matching elements **1120**, the form of the array or other pattern according to which the artificial dielectric matching elements **1120** are disposed across the substrate **1116**, and the thickness of the substrate **1116** can all be selected in consideration of the frequencies at which the transmission of radio frequency signals are desired, the range of angles at which the radio frequency signals will intersect a radome **112** or other structure incorporating the unitary frequency selective structure **1104**, and desired protection against electromagnetic interference.

[0045] As an example, but without limitation, a unitary frequency selective structure **1104** can be injection molded from a polycarbonate having a dielectric constant of 3.85. The area of the unitary frequency selective structure **1104** constituting the substrate **1116** can have a thickness of about 0.6 inch. A distance **1133** from an end **1111** of a first artificial dielectric matching element **1120** on a first side of the substrate **1116** to an end **1111** of a second artificial dielectric matching element **1120** on a second side of the substrate **1116** and aligned with the first artificial dielectric matching element **1120** may be about 1.6 inches. An exterior diameter **1123** of the artificial dielectric matching elements **1120** may be about 0.355 inches.

[0046] A unitary frequency selective structure **1104** as disclosed herein allows the construction of a radome **112** or other structure having excellent transmission performance with respect to radio frequency energy within a selected band of frequencies, and over a range of angles. The performance of the unitary frequency selective structure **1104** may differ from that of embodiments including one or more frequency selective surfaces. For example, the range of angles over which radio frequency losses remain low may be reduced as compared to embodiments incorporating a frequency selective surface. This narrowing of the angular range over which the unitary frequency selective structure **1104** is essentially transparent to radio frequency energy may be more apparent for a particular polarization than for another polarization. As a further example, the rejection of selected wavelengths by a unitary frequency selective structure **1104** may be diminished as compared to embodiments incorporating a frequency selective structure.

[0047] FIG. **13A** is a cross section in elevation and FIG. **14** is a plan view of a frequency selective structure **1304** incorporating a substrate **1316** in the form of a thick frequency selective surface **1312**. In the frequency selective structure **1304**, a pair of artificial dielectric matching layers **1308a** and **1308b** are formed by a shared set of artificial dielectric matching components **1335**. Each of the artificial dielectric matching components **1335** includes a pair of artificial dielectric matching elements **1320** disposed on opposite sides of a connecting section **1337**. In accordance with at least some embodiments of the present disclosure, the artificial dielectric matching elements **1320** and the connecting section **1337** of any one of the artificial dielectric matching components **1335** are integrally formed from the same piece of material. The thick frequency selective surface **1312** can be configured as a plate or substrate **1316** of electrically conductive material with an array of through holes or apertures **1339** formed therein. The thickness and other parameters of the thick frequency selective surface **1312** can be determined using the same or similar techniques as would be used to design a conventional thick frequency selective surface.

[0048] The thick frequency selective surface **1312** of embodiments of the present disclosure is different from a conventional thick frequency selective surface in that artificial dielectric matching components **1335** as disclosed herein are disposed within the through holes **1339**. In accordance with embodiments of the present disclosure, the connecting section **1337** of each artificial dielectric matching component **1335** is centered in the thick frequency selective surface **1312**, with a first one of the artificial dielectric matching elements **1320** disposed on a first surface side of the thick

frequency selective surface **1312**, and a second one of the artificial dielectric matching elements **1320** disposed on a second surface side of the thick frequency selective surface **1312**. The artificial dielectric matching elements **1320** included in an artificial dielectric matching component **1335** can be identical to one another, and can include a base portion **1305** extending from the connecting section **1337** and defining a closed end **1321** of an interior volume **1313**, a first interior volume portion **1317** having a first diameter **1315**, and a second interior volume portion **1325** having a second diameter **1327** that is larger than the first diameter **1315**. As an example, but without limitation, the material of an artificial dielectric matching component **1335** can be a glass, such as a borosilicate glass having a dielectric constant of 4.4. As illustrated in FIG. **13B**, a coating **1324** can be applied over the artificial dielectric matching elements **1320** of one or both artificial dielectric matching layers **1308**, filling the interior volumes **1313** of and the spaces between the artificial dielectric matching elements **1320**, to form a smooth outer surface. As in other embodiments, examples of the coating **1324** can include a dielectric material such as a low density ablator or insulator in the form of a filled elastomeric silicone, a white silicon based paint, a thick film coating, or an epoxy coating.

[0049] In accordance with embodiments of the present disclosure, the through holes **1306** can be formed by drilling holes in a plate of electrically conductive material provided as the substrate **1316** having a diameter matching an exterior diameter **1323** of the artificial dielectric matching components **1335**. The artificial dielectric matching components **1335** can be formed by molding a glass or other dielectric material. As an example, but without limitation, the thick frequency selective surface **1312** can have a thickness of about 0.111 inches. A distance **1333** from an end **1311** of the first artificial dielectric matching element **1320** of an artificial dielectric matching component **1335** to an end **1311** of the second artificial dielectric matching element **1320** of the artificial dielectric matching component **1335** can be about 0.340 inches. An exterior diameter **1323** of the artificial dielectric matching elements **1320** may be about 0.1088 inches. The diameter of the connecting section **1337** of an artificial dielectric matching component **1335** can be the same as the exterior diameter **1323** of each of the artificial dielectric matching elements **1320** of that artificial dielectric matching component **1335**. The diameter **1315** of the first interior volume portion **1317** may be about 0.0032 inches, and the diameter **1327** of the second interior volume portion **1325** may be about 0.0878 inches.

[0050] FIG. **15** is a graph depicting the transmission performance of a frequency selective structure **604**, **1104**, or **1304** incorporating artificial dielectric matching layers **608**, **1108**, or **1308** in accordance with embodiments of the present disclosure. As depicted, the normal pass band performance and the normal stop band performance of the frequency selective structure **604**, **1104**, or **1304** can be well aligned with the off-normal pass band performance and the off-normal stop band performance of the frequency selective structure **604**, **1104**, or **1304**.

[0051] FIG. **16** is a flow chart illustrating aspects of a method for forming a frequency selective structure **604** or **1304** incorporating artificial dielectric matching layers **608** or **1308** and a frequency selective surface layer **612** or **1312** in accordance with embodiments of the present disclosure. Initially, a frequency selective surface layer or surface layers **612**, or a thick frequency selective surface **1312**, is provided (step **1604**). The frequency selective surface layer or layers **612** or the thick frequency selective surface **1312** can be configured to provide desired pass band and stop band characteristics for a selected radio frequency or range of radio frequencies. The frequency selective surface layers **612** or **1312** can be designed using conventional techniques. Accordingly, a pair of frequency selective surface layers **612** can be produced as arrays **812** of open or non-conductive slots **816** disposed in unit cells **808** incorporating a metalized or otherwise conductive layer that are disposed on opposite surfaces of a substrate **616** in the form of a layer of dielectric material. A thick frequency selective surface **1312** can be produced by forming an array of through holes or apertures **1339** in a substrate **1316** in the form of a plate of conductive material.

[0052] Next, artificial dielectric matching elements **620** or **1320** are formed (step **1608**). The

artificial dielectric matching elements **620** or **1320** can be configured as cylindrical structures formed from a dielectric material. In embodiments in which an array of through holes **1339** is provided in a substrate **1316**, the artificial dielectric matching elements **1320** can be formed in pairs as part of artificial dielectric matching components **1335**. Each artificial dielectric matching element **620** or **1320** can include a closed base portion **905** or **1305** and an open distal portion **909** or **1309**. A diameter of an interior volume **913** or **1313** proximate to the base portion **905** or **1305** (i.e. of a first interior volume portion **917** or **1317**) can be smaller than a diameter of the interior volume **913** or **1313** at the open distal portion **909** or **1309** (i.e. of a second interior volume portion **925** or **1325**). The outer diameter, height, and inner diameter or diameters of each artificial dielectric matching element **620** or **1320** is based on desired the pass band and stop band of the frequency selective structure **604** or **1304**. In accordance with embodiments of the present disclosure, each artificial dielectric matching element **620** is formed as a unitary element or structure from a non-conductive material having any dielectric constant. In accordance with embodiments incorporating a thick frequency selective surface **1312**, each artificial dielectric matching element **1320** can be formed as one of a pair of artificial dielectric matching elements **1320** joined to one another by a connecting section **1337**. As examples, but without limitation, the artificial dielectric matching elements **620** or **1320** can be formed by molding, machining, or using additive manufacturing processes. In accordance with further embodiments of the present disclosure, the number of provided artificial dielectric matching elements **620** or **1320** included in the frequency selective structure **604** or **1304** is equal to the number of unit cells **808** of the frequency selective surface layers **612** or the number of through holes **1339** in the thick frequency selective surface **1312**.

[0053] Artificial dielectric matching layers **608** or **1308** are then formed by connecting the artificial dielectric matching elements **620** or **1320** to the frequency selective surface layer **612** or thick frequency selective surface **1312** (step **1612**). For example, one artificial dielectric matching element **620** can be disposed on or adjacent to each unit cell **808** of the frequency selective surface layers **612**. More particularly, the artificial dielectric matching elements **620** are disposed on or adjacent to sides of the frequency selective surface layers **612** opposite the sides of those frequency selective surface layers **612** that are facing the substrate **616** (in this example a dielectric layer). The dielectric matching elements **620** can be directly bonded or otherwise attached to an area of a corresponding unit cell **808** of a frequency selective surface layer **612** in a pick and place operation. In accordance with further embodiments of the present disclosure, an adhesive layer or other intermediate layer or layers can be interposed between the artificial dielectric matching elements **620** and the frequency selective surface layers **612**. In such an embodiment, a set or array of artificial dielectric matching elements **620** can be joined to an intermediate layer to form an artificial dielectric matching layer **608**, and that dielectric matching layer **608** can then be aligned with a frequency selective surface layer **612** so that each artificial dielectric matching element **620** included in the artificial dielectric matching layer **608** is centered in a corresponding unit cell **808** of the frequency selective surface **612**, and the dielectric matching layer **608** can be joined to the frequency selective surface layer **612**. Alternatively, an intermediate layer can first be placed on the frequency selective surface layer **612**, and the artificial dielectric matching elements **620** can then be interconnected to the intermediate layer. As another example, an artificial dielectric matching component **1335** with a pair of integral artificial dielectric matching elements **1320** joined to one another by a connecting section **1337** can be placed in each through hole **1339** in the thick frequency selective surface **1312**. Each artificial dielectric matching component **1335** can be placed such that the included artificial dielectric matching elements **1320** extend the same amount from the nearest surface of the thick frequency selective surface **1312**. Moreover, each artificial dielectric matching component **1335** can be bonded, friction fit, or otherwise secured in a corresponding through hole **1339**.

[0054] A coating **624** or **1324** can then be applied to the artificial dielectric matching layers **608** or

1308, over the artificial dielectric matching elements **620** or **1320**, to provide a uniform surface (step **1616**). The coating **624** or **1324** can be applied like a paint that is sprayed over the artificial dielectric matching layers **608** or **1308** or as an ablative. In accordance with at least some embodiments of the present disclosure, a coating **624** or **1324** can be applied over one surface of the frequency selective structure **604** or **1304**. For example, a coating **624** or **1324** may be applied over a surface of the frequency selective structure **604** or **1304** that will correspond to an exterior of a radome **112**, but omitted from a surface of the frequency selective structure **604** or **1304** that will correspond to an interior of the radome **112**. In accordance with still other embodiments, a coating **624** or **1324** can be applied across portions of a surface of a frequency selective structure **604** or **1304**, rather than across the entire surface. In accordance with still other embodiments, a coating can be omitted entirely. The resulting frequency selective structure **604** or **1304** can then be provided as a planar panel, or as a curved or otherwise formed component, and deployed as a radome or cover **112**. The process can then end.

[0055] A unitary frequency selective structure **1104** can be formed from a single piece of dielectric material. For example, a unitary frequency selective structure **1104** can be formed by injection molding plastic, such as polycarbonate. As another example, a unitary frequency selective structure **1104** can be 3D printed from PLA or ABS.

[0056] Embodiments of the present disclosure can include a single artificial dielectric matching layer **608**, **1108**, or **1308** on one side of a substrate **616**, **1116**, or **1316**. Embodiments of frequency selective structures **604** and **1304** as disclosed herein can include a single frequency selective surface layer **612** or thick frequency selective surface **1312** and a single artificial dielectric matching layer **608** or **1308**. Variations in the configuration of the artificial dielectric matching elements **620**, **920**, or **1320** within an artificial dielectric matching layer **608**, **1108**, or **1308** are also possible. For instance, the configuration and/or number of interior volume portions formed in an artificial dielectric matching element **620**, **920**, or **1320** can be selected based on desired performance characteristics. As another example, the configuration of the artificial dielectric matching elements **620**, **920**, or **1320** can vary based on location within the frequency selective structure **604**, **904**, or **1304**. In accordance with still other embodiments of the present disclosure, a size of the unit cells **808**, a spacing of adjacent artificial dielectric matching elements **620**, **920**, or **1320**, the size of the artificial dielectric matching elements **620**, **920**, or **1320**, or the configuration of the artificial dielectric matching elements **620**, **920**, or **1320** can vary within a frequency selective structure **604**, **904**, or **1304**.

[0057] The foregoing disclosure has been presented for purposes of illustration and description. Further, the description is not intended to limit the disclosure to the form disclosed herein. Consequently, variations and modifications commensurate with the above teachings, within the skill or knowledge of the relevant art, are within the scope of the present disclosure. The embodiments described hereinabove are further intended to explain the best mode presently known of practicing the disclosure and to enable others skilled in the art to utilize the disclosure in such or in other embodiments and with the various modifications required by their particular application or use of the disclosure. It is intended that the appended claims be construed to include alternative embodiments to the extent permitted by the prior art.

Claims

1. A frequency selective structure, comprising: a substrate; a first artificial dielectric matching layer, wherein the first artificial dielectric matching layer is disposed proximate a first side of the substrate, wherein the first artificial dielectric matching layer includes a first plurality of artificial dielectric matching elements, and wherein each artificial dielectric matching element in the first plurality of artificial dielectric matching elements includes: a closed base portion; an open distal portion; and an interior volume, wherein the open distal portion is adjacent an open first end of the

interior volume, and wherein the closed base portion is adjacent a closed second end of the interior volume.

2. The frequency selective structure of claim 1, further comprising: a second artificial dielectric matching layer, wherein the second artificial dielectric matching layer is disposed on a second side of the substrate, wherein the second side of the substrate is opposite the first side of the substrate, wherein the second artificial dielectric matching layer includes a second plurality of artificial dielectric matching elements, and wherein each artificial dielectric matching element in the second plurality of artificial dielectric matching elements includes: a closed base portion; an open distal portion; and an interior volume, wherein the open distal portion is adjacent an open first end of the interior volume, and wherein the closed base portion is adjacent a closed second end of the interior volume.

3. The frequency selective structure of claim 1, wherein the substrate is a dielectric layer, the frequency selective structure further comprising: a first frequency selective surface layer disposed on a first side of the dielectric layer, wherein the first frequency selective surface layer is disposed between the first artificial dielectric matching layer and the dielectric layer, and wherein the first frequency selective surface layer includes a first plurality of unit cells.

4. The frequency selective structure of claim 3, wherein the unit cells of the first plurality of unit cells are disposed in a first two-dimensional array, wherein each artificial dielectric matching element of the first plurality of artificial dielectric matching elements is disposed within an area of a corresponding unit cell of the first plurality of unit cells.

5. The frequency selective structure of claim 3, further comprising: a second frequency selective surface layer disposed on a second side of the dielectric layer, wherein the second side of the dielectric layer is opposite the first side of the dielectric layer, and wherein the second frequency selective surface layer includes a second plurality of unit cells; and a second artificial dielectric matching layer, wherein the second frequency selective surface layer is between the second artificial dielectric matching layer and the dielectric layer, wherein the second artificial dielectric matching layer includes a second plurality of artificial dielectric matching elements, and wherein each artificial dielectric matching element in the second plurality of artificial dielectric matching elements includes: a closed base portion; an open distal portion; and an interior volume, wherein the open distal portion is adjacent an open first end of the interior volume, and wherein the closed base portion is adjacent a closed second end of the interior volume.

6. The frequency selective structure of claim 5, wherein the unit cells of the first plurality of unit cells are disposed in a first two-dimensional array, wherein each artificial dielectric matching element of the first plurality of artificial dielectric matching elements is disposed within an area of a corresponding unit cell of the first plurality of unit cells, wherein the unit cells of the second plurality of unit cells are disposed in a second two-dimensional array, and wherein each artificial dielectric matching element of the second plurality of artificial dielectric matching elements is disposed within an area of a corresponding unit cell of the second plurality of unit cells.

7. The frequency selective structure of claim 2, wherein each artificial dielectric matching element in the first plurality of artificial dielectric matching elements is coupled to an artificial dielectric matching element in the second plurality of artificial dielectric matching elements by a connecting section to form a plurality of artificial dielectric matching components, wherein the substrate is a conductive material having an array of through holes, and wherein the connecting section of each of the artificial dielectric matching components is disposed in one of the through holes.

8. The frequency selective structure of claim 1, wherein each of the artificial dielectric matching elements has a cylindrical exterior form.

9. The frequency selective structure of claim 1, wherein each of the artificial dielectric matching elements has a cylindrical interior volume.

10. The frequency selective structure of claim 1, wherein the interior volume includes first and second interior volume portions.

- 11.** The frequency selective structure of claim 10, wherein the first interior volume portion has a first diameter and is proximate to the closed base portion, and wherein the second interior volume portion has a second diameter and is proximate to the open distal portion.
 - 12.** The frequency selective structure of claim 11, wherein the first diameter is smaller than the second diameter.
 - 13.** The frequency selective structure of claim 1, further comprising: a coating disposed over the artificial dielectric matching elements of the first artificial dielectric matching layer.
 - 14.** The frequency selective structure of claim 1, wherein a dielectric constant of a material of the artificial dielectric matching elements is from 1.4 to 4.0.
 - 15.** The frequency selective structure of claim 1, wherein the frequency selective structure is included in a radome.
 - 16.** A method for forming a frequency selective structure, comprising: providing a first plurality of artificial dielectric matching elements, wherein each artificial dielectric matching element in the first plurality of dielectric matching elements includes: a closed base portion; and an open distal portion; and disposing the first plurality of artificial dielectric matching elements on or adjacent a first surface of a substrate.
 - 17.** The method of claim 16, further comprising: providing a second plurality of artificial dielectric matching elements, wherein each artificial dielectric matching element in the second plurality of dielectric matching elements includes: a closed base portion; and an open distal portion; and disposing the second plurality of artificial dielectric matching elements on or adjacent a second surface of a substrate, wherein the second surface of the substrate is opposite the first surface of the substrate.
 - 18.** The method of claim 16, wherein the first plurality of artificial dielectric matching elements are disposed on or adjacent the first surface of the substrate using a pick and place operation.
 - 19.** The method of claim 16, further comprising: disposing a coating over the first plurality of artificial dielectric matching elements.
 - 20.** A radome, comprising: a frequency selective structure, including: a substrate; an artificial dielectric matching layer, wherein the artificial dielectric matching layer includes a plurality of artificial dielectric matching elements, and wherein each of the artificial dielectric matching elements includes: a closed base portion, wherein the closed base portion is disposed on or adjacent to a first surface of the substrate; an open distal portion; and an interior volume.
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